Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf____ 2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ___Serial0/0/0___ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ___129___ 2

Does R2 show up as an OSPF neighbor on R1? ____Yes____ 2

Does R2 show up as an OSPF neighbor on R3? ____No____ 2

What does this information tell you?

_It indicates that a passive interface will allow the network prefix to be advertised, but it prevents the router from forming a neighbor relationship 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ____Serial0/0/1________ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

_65, R3 outgoing interface S0/0/1 (cost 64) + R2 interface G0/0 (cost 1)____
_

Is R2 listed as an OSPF neighbor to R3? ___Yes______

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

___to allow it to accurately choose the fastest path in modern networks._________________

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Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

to 192.168.3.0/24:

781+1=782

To 192.168.23.0/30:

781+64=845

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

______1562._all serial interfaces across all routers was changed to 128 Kbpsmaking the cost = 781____

____

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

___it has the lower accumulated cost____ 4

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   __It prevents the Router ID from changing if a physical interface goes down, ensures stability during DR/BDR elections.

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2. Why is the DR/BDR election process not a concern in this lab?

   _____because it only occur on multi-access networks.

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3. Why would you want to set an OSPF interface to passive?

   _____it allows the router to advertise that network segment into OSPF without sending OSPF Hello packets out of that specific interface.

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4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   _____Areas.____ ______

   4

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5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   ______network 10.0.0.0 0.255.255.255 area 0.

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6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   ______EIGRP.____ _______

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